

UniDex: A Robot Foundation Suite for Universal Dexterous Hand Control from Egocentric Human Videos

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<https://unidex-ai.github.io/>

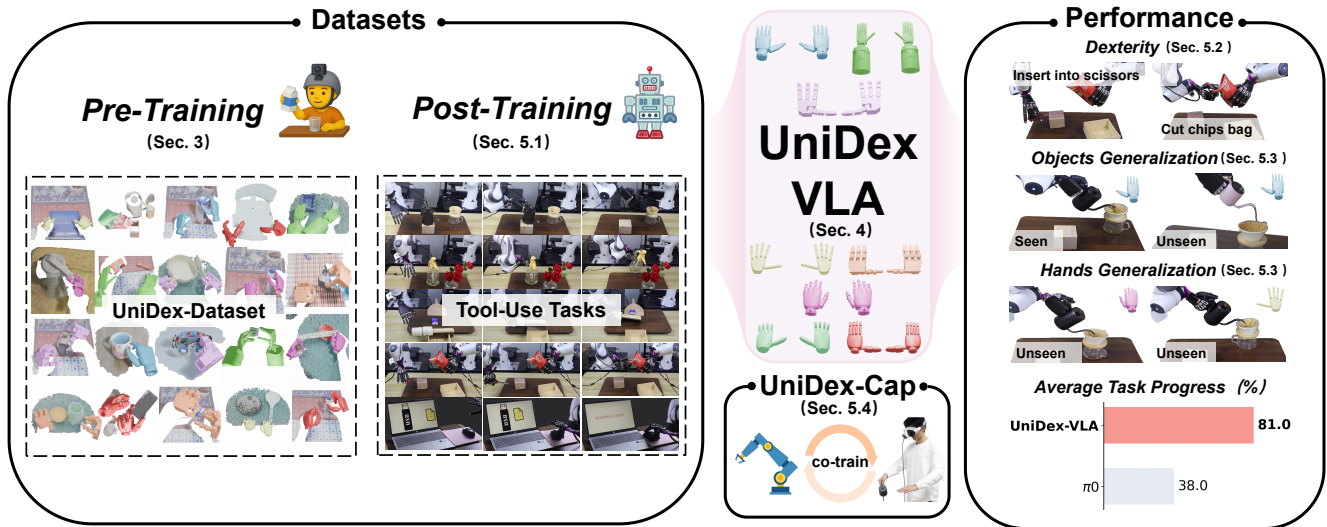


Figure 1. We introduce **UniDex**, a robot foundation suite for heterogeneous dexterous hand embodiments. We first curate UniDex-Dataset from egocentric human videos to obtain a diverse, robot-centric dataset for large-scale pretraining. Building on this, we train UniDex-VLA, a unified 3D VLA model that is finetuned with task demonstrations and evaluated on challenging real-world tool-use tasks. The policy exhibits strong dexterous capabilities, zero-shot object and cross-hand generalization and significantly outperforming existing VLA baselines. In addition, we design a practical setup, UniDex-Cap to support human-robot data co-training, further reducing the data cost.

Abstract

Dexterous manipulation remains challenging due to the cost of collecting real-robot teleoperation data, the heterogeneity of hand embodiments, and the high dimensionality of control. We present UniDex, a robot foundation suite that couples a large-scale robot-centric dataset with a unified vision-language-action (VLA) policy and a practical human-data capture setup for universal dexterous hand control. **First**, we construct UniDex-Dataset, a robot-centric dataset over 50K trajectories across eight dexterous hands (6–24 DoFs), derived from egocentric human video datasets. To transform human data into robot-

executable trajectories, we employ a human-in-the-loop re-targeting procedure to align fingertip trajectories while preserving plausible hand-object contacts, and we operate on explicit 3D pointclouds with human hands masked to narrow kinematic and visual gaps. **Second**, we introduce the Function-Actuator-Aligned Space (FAAS), a unified action space that maps functionally similar actuators to shared coordinates, enabling cross-hand transfer. Leveraging FAAS as the action parameterization, we train UniDex-VLA, a 3D VLA policy pretrained on UniDex-Dataset and finetuned with task demonstrations. **In addition**, we build UniDex-Cap, a simple portable capture setup that records synchronized RGB-D streams and human hand poses and con-

verts them into robot-executable trajectories to enable human–robot data co-training that reduces reliance on costly robot demonstrations. On challenging tool-use tasks across two different hands, UniDex-VLA achieves 81% average task progress and outperforms prior VLA baselines by a large margin, while exhibiting strong spatial, object, and zero-shot cross-hand generalization. Together, UniDex-Dataset, UniDex-VLA, and UniDex-Cap provide a scalable foundation suite for universal dexterous manipulation.

1. Introduction

In recent years, learning from demonstrations [6, 7, 13, 27, 58, 67, 74] has become the de facto paradigm for visuomotor control, enabling robots to acquire complex skills and motion patterns. However, achieving general, human-level manipulation under supervised learning remains challenging. Collecting real-robot demonstrations is labor-intensive and scales poorly, creating a persistent data bottleneck. Moreover, most robot foundation policies focus on parallel-jaw grippers, while foundation models for dexterous hands remain scarce—even though everyday tool-use often requires dexterous hands and many tasks (e.g., using scissors or spray bottles) are infeasible with grippers.

Simply porting gripper-based VLA designs to dexterous hands is insufficient. Building foundation models for dexterous hands is substantially more challenging than for grippers. The key difficulties are: (i) dexterous hand data are harder to collect than gripper data, and large, broadly usable pretraining datasets remain limited; (ii) dexterous hands vary widely in DoFs, morphology, kinematics, and appearance, leading to poor transfer of data and policies across hands; and (iii) dexterous hand control is inherently high-dimensional, demanding expressive action spaces and effective learning algorithms.

To address pretraining data scarcity, we leverage the fact that dexterous robot hands are designed to mimic human hands and often share similar action patterns, while humans naturally generate abundant manipulation data in daily life. Egocentric human videos are cheaper, more diverse than robot teleoperation data and easier to scale. We therefore transform human videos into robot-executable trajectories to build a robot-centric dataset from human activity. However, there are substantial *kinematic* and *visual* gaps between human and robot hands. To close these gaps, we (i) introduce a human-in-the-loop retargeting procedure that combines fingertip-based inverse kinematics with interactive adjustment to align robot fingertip trajectories with human trajectories, ensuring physically plausible hand–object contacts; and (ii) mask the human hand in the visual stream and attach the retargeted robot hand into scene pointclouds to reduce visual mismatch.

Following this human-to-robot transformation pipeline,

we construct **UniDex-Dataset** by building on open-source egocentric RGB-D manipulation videos [4, 28, 35, 36]. **UniDex-Dataset** is a unified foundation dataset comprising 9M paired image–pointcloud–action frames and over 50K trajectories across eight dexterous hand platforms, covering active DoFs from 6 to 24. To our knowledge, UniDex-Dataset is the first dataset to span such a broad spectrum of dexterous hand morphologies at this scale. We also provide protocols that allow researchers to contribute new hands or human datasets with minimal effort, continually scaling UniDex-Dataset and accelerating progress on dexterous manipulation.

To tackle heterogeneous embodiments and high-dimensional control, we further define a unified action space, the **Function–Actuator–Aligned Space (FAAS)**, which maps functionally similar actuators to shared coordinates. FAAS provides a function-centric control interface and enables skill transfer across different hands. Building on FAAS, we train **UniDex-VLA**, a 3D vision–language–action policy pretrained on UniDex-Dataset and finetuned with task demonstrations, serving as a foundation model that supports diverse dexterous hands.

In addition, we design a portable human-data capture setup, **UniDex-Cap**, which records synchronized RGB-D streams and human hand poses and converts them into robot-centric trajectories via the same transformation pipeline. UniDex-Cap enables efficient co-training on transformed human data together with smaller amounts of robot data, reducing teleoperation cost while preserving performance.

We evaluate UniDex-VLA on five challenging real-world tool-use tasks across two different hands. Across these tasks, **UniDex-VLA** achieves strong performance, outperforming other VLA baselines by a large margin (e.g., 81% average task progress vs. π_0 [7] at 38%), and demonstrates strong spatial, object, and cross-hand generalization; with FAAS and pretraining, it transfers skills to unseen hands in a zero-shot manner. Leveraging UniDex-Cap, we also provide a quantitative study showing how transformed human data can reduce post-training costs via human–robot co-training.

Our contributions are summarized as follows:

- **UniDex-Dataset:** a unified, diverse dexterous hand dataset (9M paired frames, over 50K trajectories, 8 hands, 6–24 DoFs) that supports large-scale pretraining toward universal dexterous hand foundation models.
- **FAAS & UniDex-VLA:** a function–actuator–aligned unified action space and a pretrained 3D vision–language–action model that achieves state-of-the-art performance on real-robot benchmarks, with strong spatial, object, and cross-hand generalization.
- **Human–Robot Data Co-training with UniDex-Cap:** a simple portable capture setup and pipeline that support

human–robot data co-training; we quantitatively study how transformed human data can partially substitute real-robot demonstrations during post-training, showing that egocentric human videos both scale pretraining and reduce real-robot data needs.

2. Related Work

2.1. Dexterous Manipulation

Early research on dexterous manipulation was grounded in analytic and classical control formulations [2, 3, 26, 40, 43], and has since progressed toward learning-based methods that enable in-hand reorientation, rotation, and grasping [1, 10, 17, 19, 22, 24, 31, 45, 50, 54, 62, 65, 70, 75, 76]. Despite these advances, most approaches are tailored to specific tasks (grasping) or hardware and struggle to generalize to everyday tool-use. In contrast, we present UniDex-VLA, a foundation model aimed at general-purpose dexterous hand control.

2.2. Robot Foundation Policies and Unified Action Space

Diffusion-based policies and their variants constitute strong imitation-learning baselines [13, 51, 57, 58, 67]. With the rise of LLMs and VLMs, vision–language–action (VLA) models [6–8, 23, 27, 33, 39, 71–73] further scale imitation learning, but most existing approaches are pretrained on large-scale gripper-centric datasets. Recent efforts toward dexterous VLAs [22, 75] leverage simulation or limited real-world data, typically focusing on grasping and relying on hand-specific representations. In contrast, UniDex-VLA is pretrained on UniDex-Dataset to serve as a unified foundation policy for more general dexterous manipulation.

Designing a unified action space for robot foundation policies to handle embodiment heterogeneity is crucial for cross-embodiment generalization. RDT-1B [33] preserves the semantic structure of control signals, while π_0 [7] adopts a left-aligned action representation, and other methods introduce latent action spaces [8, 71]. However, these approaches primarily target gripper-centric actions. EgoVLA [60] attempts to leverage human parameters as a dexterous representation, but requires inverse kinematics in the post-training stage, which introduces additional errors, particularly for high-DoF dexterous hands. In contrast, FAAS provides a function-centric unified action representation that is post-processing-free, enabling more reliable cross-hand skill transfer.

2.3. Learning from Human Videos

Learning from human videos mitigates the data cost bottleneck but introduces visual and kinematic domain gaps. Prior work uses human hand trajectories for planning or control [9, 29, 34, 46, 52, 55, 63]; others apply retarget-

ing with sim-to-real pipelines [11, 30, 66] or human-in-the-loop corrections [53], and some co-train with robot data [25, 48, 64, 78] to bridge the gap. However, many such pipelines primarily target grippers or do not scale robustly. There are also approaches that pretrained on egocentric human videos without explicit supervision of hand motion [41, 42, 61, 68]. More recent methods pretrain foundation models on egocentric videos to predict human hand motion, followed by specialized post-training to align with robot actions [38, 60], however these additional alignment stages can be complex and brittle. Our approach instead generates robot-centric dexterous hand supervision for pre-training, removing the need for specialized alignment tricks during fine-tuning while maintaining cross-hand control.

3. UniDex-Dataset

3.1. Overview

UniDex-Dataset is derived from four RGB-D egocentric human-manipulation datasets—H2O [28], HOI4D [35], HOT3D [4], and TACO [36]. We annotate language instructions if needed, segment videos into trajectory clips aligned with those instructions, and filter out invalid segments.

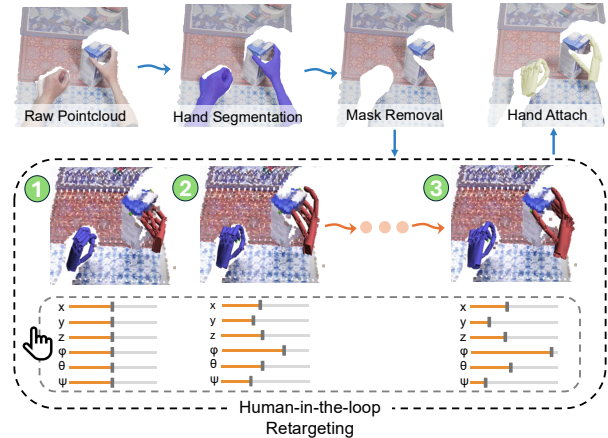


Figure 2. The figure illustrates the complete human–robot transformation pipeline. Starting from the raw scene pointcloud, we first mask out the human hands. We then perform human-in-the-loop retargeting through a user-friendly GUI in which the user only needs to **adjust slider bars** to modify the dummy base offset. ① shows the retargeted result without adjustment, whereas ③ shows the final configuration with improved, more plausible hand–object contact. Finally, after kinematic retargeting, we attach the retargeted robot dexterous hands to the scene.

The transformation from human data to robot-executable trajectories is illustrated in Fig. 2 and detailed in the next subsection. Applying this pipeline, we construct **UniDex-Dataset** comprising 9M paired image–pointcloud–action frames (recorded at 30 fps) and over 50k trajectories across eight dexterous hand platforms (Inspire, Leap, Shadow, Allegro, Ability, Oymotion, Xhand, and Wuji), covering ac-

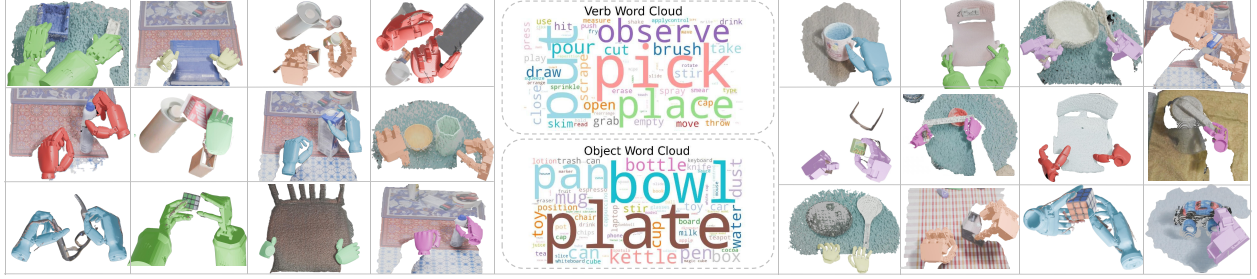


Figure 3. **UniDex-Dataset visualization.** We show a verb-object word cloud and a subset of UniDex-Dataset. **Colors** denote **different hands** (arbitrarily assigned; black corresponds to pretraining data). UniDex-Dataset spans diverse everyday tasks across a wide range of dexterous hand embodiments, including using a mobile phone, opening a milk carton, stir-frying with a spatula, lifting a chair, solving a Rubik’s cube, and more.

tive DoF from 6 to 24. Figure 3 visualizes the verb-object word cloud for the dataset and a subset of the data, spanning diverse daily manipulation tasks such as using a mobile phone, opening a milk carton, and stir-frying with a spatula. Table 1 compares UniDex-Dataset with released collected dexterous manipulation datasets [20, 37, 56] along the axes of trajectory count, hand variety and scene diversity, and supported perception modalities, highlighting the advantages of UniDex-Dataset. Owing to its diversity and robot-centric formulation—i.e., with minimal embodiment gap to the post-training stage—UniDex-Dataset serves as a strong foundation for pretraining dexterous manipulation models.

3.2. Human-Robot Transformation

Transforming human data into robot trajectories requires overcoming two core gaps: *kinematic* and *visual*. We outline our methods below.

3.2.1. Kinematic Retargeting

Fingertips are the primary contact points in human-object interaction. Our goal is to align human fingertip trajectories with those of the robot hand in 3D, while allowing a global hand-base adjustment to better ensure physically plausible contact.

Given a human hand pose, we extract m fingertip targets

$$X^* = [x_1^*, \dots, x_m^*] \in \mathbb{R}^{3 \times m}, \quad (1)$$

where m equals the number of robot fingers. The global human hand transform in the world frame is T_{hand} .

To precisely apply fingertip-based IK while permitting a base adjustment, we introduce a *6-DoF alignment offset*, implemented as a **dummy base** inserted before the real robot base. Let T_{offset} be the rigid transform from the dummy base to the real base, and let $T_{\text{world}}^{\text{dummy}}$ be the dummy-base pose in the world frame. The forward kinematics of fingertip i is

$$x_i(q; T_{\text{offset}}) = \text{Trans}\left(T_{\text{world}}^{\text{dummy}} T_{\text{offset}} T_i(q)\right) \in \mathbb{R}^3, \quad (2)$$

where $T_i(q)$ is the homogeneous transform from the robot base to fingertip i , and $\text{Trans}(\cdot)$ extracts the translation. We set $T_{\text{world}}^{\text{dummy}} = T_{\text{hand}}$ and keep it fixed during optimization. Stacking fingertip residuals yields the IK error:

$$e(q, T_{\text{offset}}) = \begin{bmatrix} x_1(q; T_{\text{offset}}) - x_1^* \\ \vdots \\ x_m(q; T_{\text{offset}}) - x_m^* \end{bmatrix} \in \mathbb{R}^{3m}. \quad (3)$$

For robot hands containing *mimic joint structures* (e.g., Inspire, Oymotion, Agility), we handle dependent joints through an iterative correction process. After solving the primary IK problem, each mimic joint j_s is updated from its master joint j_m as

$$q_{j_s} = k q_{j_m} + c \quad (4)$$

consistent with the kinematic model specification, where k and c denote the mimic constraints. This correction is repeated for N iterations, re-evaluating fingertip error each time until convergence.

For implementation, we provide a user-friendly and rapid process. The whole pipeline is a two-stage, human-in-the-loop retargeting procedure.

1. **Automatic stage.** Given an initial T_{offset} , we solve Eq. 3 via PyBullet [15]’s multi-end-effector IK solver to obtain a joint configuration q that minimizes fingertip error while satisfying joint limits and damping.
2. **Interactive stage.** A lightweight GUI exposes the six degrees of freedom of T_{offset} (three translations and three rotations, as shown in Fig. 2) and other configuration for IK solver. The user visually inspects alignment and manually adjusts T_{offset} ; after each adjustment, we re-solve the IK problem. This process typically converges within a few manual tweaks, producing robust fingertip alignment across diverse poses. ① and ③ in Fig. 2 shows the comparison between and after the interactive stage.

For each human dataset and each dexterous hand, we perform a basic interactive calibration to select dummy base offsets to handle systematic differences across datasets

Dataset	# of Trajectories	# of Hands	Language Annotations	Varied Scenes	RGB	Depth	Pointcloud
UniDex-Dataset	52K	8	✓	✓	✓	✓	✓
ActionNet [20]	30K	2	✓	✗	✓	✓	✗
RoboMind [56]	19K	1	✓	✗	✓	✓	✗
RealDex [37]	2K	2	✓	✗	✓	✓	✓

Table 1. **Comparison between UniDex-Dataset and other dexterous manipulation datasets.** UniDex-Dataset advances in total trajectories, variety across hands/actions/scenes, and supports for all perception modalities. ✗denotes the pointcloud in ActionNet [20] is very low-quality.

(e.g., coordinate frames/ hand-pose estimation bias) and hand morphology differences. We then adjust a small subset of frames, focusing on contact-rich segments to improve contact plausibility. In practice, we find the basic calibration suffices to cover the vast majority of trajectories, enabling our transformation pipeline to scale to large egocentric datasets with modest human effort.

3.2.2. Visual Alignment

We compute pointclouds from RGB-D frames. Then to reduce the visual gap, we mask human hands (using WiLoR [44] together with SAM2 [49]) and remove the corresponding points. We then place the retargeted robot-hand mesh into the scene and render its geometry into the pointcloud. Finally, we reproject the fused pointcloud back to the RGB-D frame via a pinhole camera model [21] to avoid occlusions caused by incorrect depth ordering, matching the single-view setting used during real-world fine-tuning.

4. UniDex-VLA

4.1. Unified Action Space: FAAS

We pretrain our robot foundation model on UniDex-Dataset, which spans diverse dexterous-hand embodiments. A unified action space that enables transfer across hands is therefore critical. To this end, we introduce a simple yet effective action representation, the **Function–Actuator–Aligned Space (FAAS)**. For any dexterous hand with n actuated DoFs in its kinematic model, each *actuator* is mapped to the FAAS *index* corresponding to its functional role. Here we use “actuator” broadly to denote any controllable DoF/channel derived from the robot URDF, including mimic joints when present.

Conceptually, FAAS exposes a function-centric control interface shared across embodiments rather than a URDF-specific joint space. Although dexterous hands differ in link lengths, couplings, and layouts, they all implement a small set of functional primitives—such as thumb–index pinch, finger curling around handles, or lateral ab-/adduction for stabilization. FAAS groups actuators by these functional roles and maps them into a common coordinate system, discarding embodiment-specific nuisance factors while preserving task-relevant control semantics. Figure 4 illustrates, for the thumb and ring fingers of different hands, how indi-

vidual joints are mapped to FAAS indices. FAAS further encodes wrist poses in the first 18 dimensions (9 per hand), and allocates 32 slots for each hand’s finger actuators, yielding an 82-dimensional action vector. The details of joint mapping for different hands are shown in Sec. C and Fig. 15 in Appendix.



Figure 4. **Function–Actuator–Aligned Space (FAAS).** We show the thumb and ring fingers of Oymotion (11 actuators), Allegro (16), Inspire (12), and Wuji (20), with colors denoting individual joints, curves indicating rotation directions, and dotted lines indicating rotation axes. Indices {0,1,3,5,6} are aligned across all four hands because the corresponding joints share similar functional roles.

4.2. VLA Policy

UniDex-VLA aims to be a 3D, language-conditioned foundation model for dexterous control. Unlike prior VLAs that pair 2D encoders with low-dimensional gripper actions, our setting is inherently volumetric and high-DoF: tool-use requires reasoning about fine 3D geometry and contact affordances, especially in the egocentric single-view observation. By coupling 3D visual inputs with the unified FAAS action space, UniDex-VLA aligns geometric perception and control in a shared representation, supporting spatial, object, and cross-hand generalization.

4.2.1. Observations and Action Outputs

As shown in Fig. 5, the observation at time t is $o_t = [P_t, \ell_t, q_t]$, where P_t is a single-view colored pointcloud derived from an RGB-D image and then cropped and down-sampled, ℓ_t is a natural-language instruction, and q_t is a vector of robot proprioceptive states. We model $p(A_t | o_t)$, where $A_t = [a_t, \dots, a_{t+H-1}]$ denotes an H -step action chunk [74]. Both q_t and each a_t are represented in FAAS.

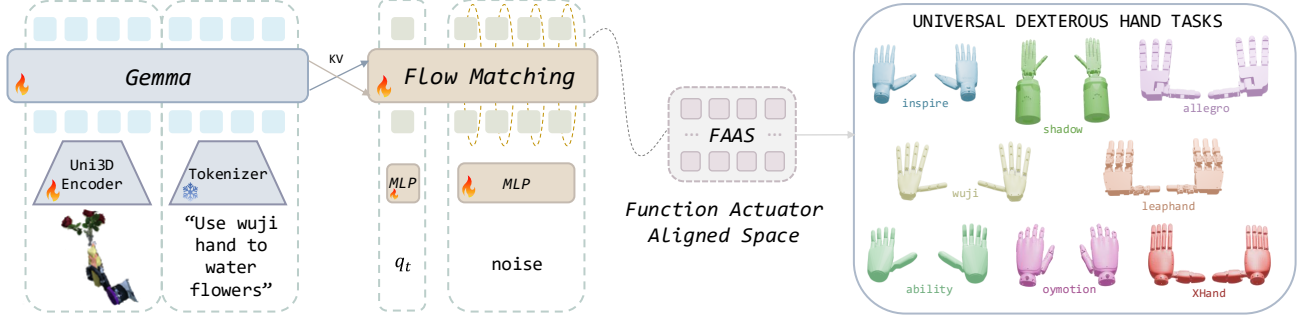


Figure 5. **Overview of UniDex-VLA.** At time t , the model consumes a single-view colored pointcloud P_t , a language instruction ℓ_t , and proprioception q_t , and predicts an H -step action chunk $A_t = [a_t, \dots, a_{t+H-1}]$ expressed in the unified action space FAAS. Uni3D [77] encodes the colored pointcloud; features are fused with text and proprioception in the backbone and decoded into FAAS actions. The policy is pretrained on UniDex-Dataset and optimized with a conditional flow-matching objective.

For the wrist in q_t , we use an *absolute* pose; for action outputs, we adopt a *relative* wrist pose with respect to the first frame of the action chunk, following UMI [14]. For dexterous-hand joints, we likewise use abstracted representations in both q_t and a_t .

4.2.2. Model Architecture

The UniDex-VLA architecture largely follows π_0 [7], with modifications for pointcloud inputs. Specifically, we replace the SigLIP [69] 2D vision encoder in PaliGemma [5] with Uni3D [77], a strong 3D pointcloud encoder. Uni3D adopts a vanilla ViT [18] design and is initialized from a 2D pretrained ViT, aligning pointcloud features with image-text-aligned features. We train the policy with a conditional flow-matching objective and generate denoised action chunks at inference time via forward-Euler integration [32]. More details of UniDex-VLA training are shown in Sec. A in Appendix.

5. Experiments

5.1. Experimental Setup

Hardware Platform. Our real-world experiments use a 7-DoF Franka robotic arm equipped with three dexterous end-effectors: an Inspire Hand (6 active, 12 full DoFs), a Wuji Hand (20 active DoFs), and an Oymotion Hand (6 active, 11 full DoFs), all mounted at the end-effector. An Intel RealSense L515 provides egocentric RGB-D observations for all experiments. The complete workstation is shown in Fig. 6.

Task Description. Everyday manipulation commonly involves many tools designed for human hands—e.g., scissors, spray bottles, and sweepers—which impose stringent requirements on finger coordination and in-hand reconfiguration. To better assess the dexterity and generality of our approach, we evaluate five challenging tool-use tasks, with visualization of different stages in Fig. 7: (i) **Make Coffee** (Inspire Hand): Grasp the kettle and lift it to the dripper to pour water to make pour-over coffee. Task decomposed

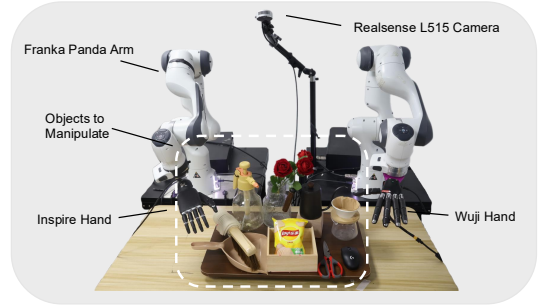


Figure 6. Real-world experiments setup overview

into kettle grasping (**Grasp**) and water pouring (**Pour**). (ii) **Sweep Objects** (Inspire Hand): Grasp a sweeper and sweep tabletop objects into a dustpan. Task decomposed into sweeper grasping (**Grasp**) and sweeping (**Sweep**). (iii) **Water Flowers** (Wuji Hand): Grasp a spray bottle, lift it, and press the trigger with the thumb to water flowers. Task decomposed into bottle grasping (**Grasp**) and pressing trigger to water (**Press**). (iv) **Cut Bags** (Wuji Hand): Insert thumb, middle and ring fingers into scissors and grasp them in a human-like manner to cut bags. Task decomposed into scissors grasping (**Grasp**) and cutting (**Cut**). (v) **Use Mouse** (Wuji Hand): Place fingers on a computer mouse and use it to drag a file into a USB folder in the desktop interface and click the mouse to finish. We report the mean success rate across all task stages as the **average task progress**, which serves as our primary metric for comparing methods.

Demonstration Collection. We build our teleoperation system on OpenTeleVision [12] and dex-retargeting [47] with Apple Vision Pro. We only collect 50 demonstrations per task for fine-tuning.

Baselines. We compare **UniDex-VLA** with representative imitation learning and VLA methods: Diffusion Policy (DP) [13], 3D Diffusion Policy (DP3) [67], and the strong VLA baseline π_0 [7] pretrained on gripper action datasets. To directly assess the effect of pretraining, we include UniDex-VLA (No Pretrain). We adopt FAAS for UniDex-VLA (No Pretrain) and π_0 , and retain low-dimensional out-

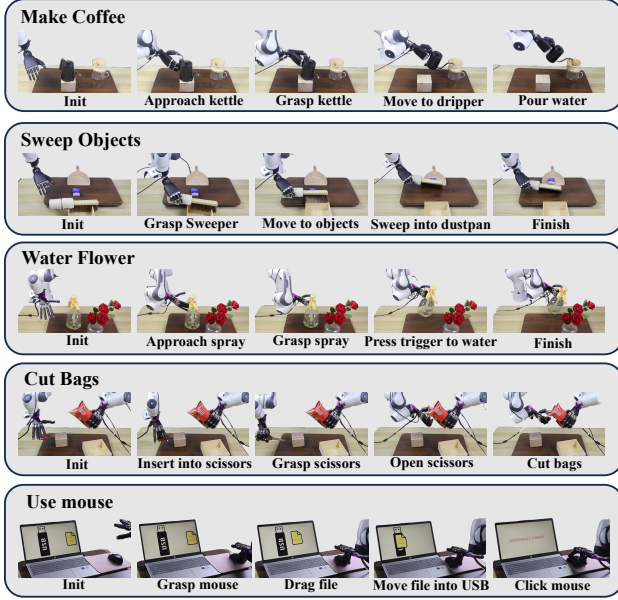


Figure 7. Our real-robot benchmark comprises 5 challenging tool-use tasks. We visualize the key stages of each task, illustrating the precise dexterous control required to successfully complete them.

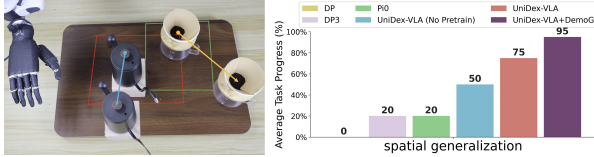


Figure 8. **Spatial generalization.** Left: the kettle and dripper are placed at *out-of-distribution* (OOD) positions relative to training demonstrations. Red and green lines circling regions denote the training placement ranges for the kettle and dripper, respectively. Right: average task progress for different methods (10 trials each).

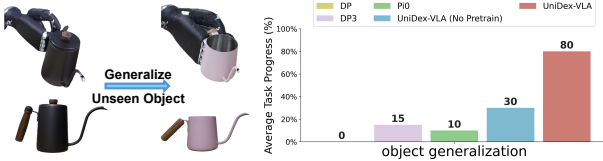


Figure 9. **Object generalization.** Left: we replace the original black kettle with a smaller purple kettle that differs in color, size, and functional parts (handle & spout). Right: average task progress for different methods (10 trials each).

5.2. Performance

We report results on five real-world manipulation tasks across two dexterous hands at Fig. 11. The results show that, with only 50 demonstrations per task, **UniDex-VLA** attains high success rates on these challenging, long-horizon tool-use tasks and surpasses all baselines by a large margin, including on the especially difficult *Use Scissors to Cut Bags* task. The performance gap between UniDex-VLA (No-Pretrain) and UniDex-VLA further provides a clear

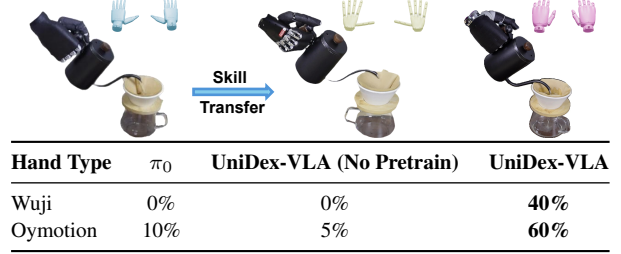


Figure 10. **Hand generalization (zero-shot skill transfer).** We transfer a policy trained on the Inspire Hand to Wuji and Oymotion. Table reports **average task progress (%)** under zero-shot deployment (10 trials each).

ablation of the benefit of pretraining on UniDex-Dataset. Computing relative improvement over the best competing method (Fig. 11), UniDex-VLA achieves the largest gain on the hardest setting, *Use Scissors to Cut Bags*, with an **84.6%** increase in average task progress. Overall, these results indicate that pretraining endows UniDex-VLA with strong motion priors for dexterous hand control, particularly on highly dexterous tool-use tasks, enabling more efficient adaptation to new and challenging behaviors.

5.3. Generalization

Beyond outperforming performance, UniDex-VLA demonstrates strong spatial, object, and hand generalization.

Spatial Generalization. UniDex-VLA benefits from 3D perception, and pointclouds further enable simple, automatic data augmentation via geometric editing. In the *Make Coffee* experiment, we segment the pointclouds of the kettle and the dripper, and translate them along the table’s x/y axes to sweep across the workspace and generate out-of-distribution (o.o.d.) placements. After editing the pointclouds, the corresponding robot states are aligned to the new scenes using Task and Motion Planning (TAMP) [16]. DemoGen [59] provides an automated pipeline for this procedure. As shown in Fig. 8, UniDex-VLA generalizes well across spatial configurations; with DemoGen [59] augmentation, it approaches very high success rate over full workspace.

Object Generalization. As in Fig. 9, we replace the black kettle with a smaller purple kettle that differs in color, size, and functional parts (handle & spout). **UniDex-VLA** maintains strong performance on this unseen object, indicating generalizable tool understanding capacity crucial for robust and general tool-use.

Hand Generalization (Skill Transfer). We evaluate cross-hand transfer by taking a policy trained to *Make Coffee* on the Inspire Hand (6 active DoF) and deploying it *zero-shot* on Wuji (20 active DoFs) and Oymotion (6 active DoFs with different kinematics). As shown in Fig. 10, **UniDex-VLA** achieves **60%** success on Oymotion and **40%** on Wuji without any fine-tuning, whereas baselines are near zero. These results highlight that pretraining across diverse dex-

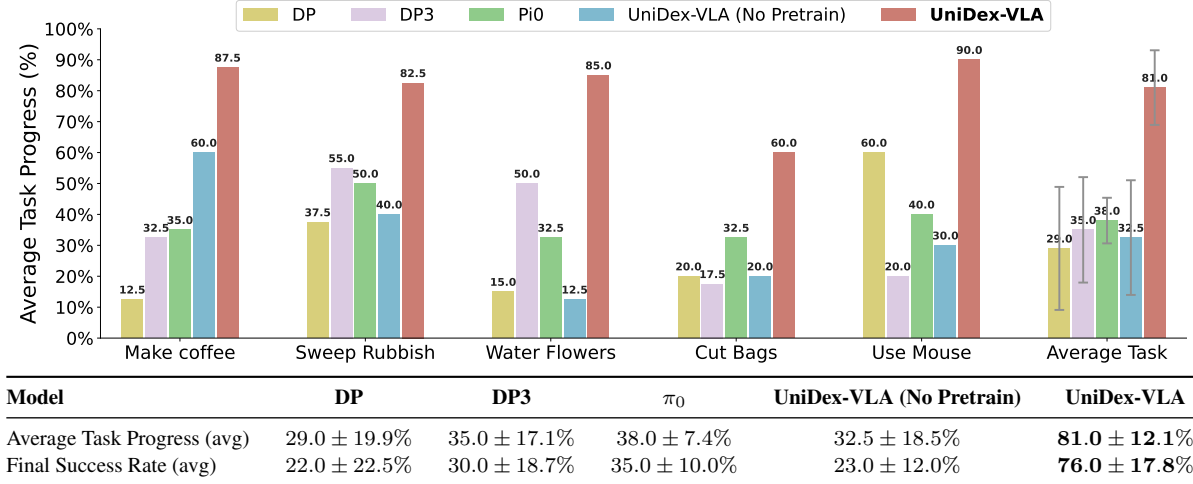


Figure 11. Average task progress across five real-world tasks (top), with aggregate averages of average task progress and final success rate (bottom) over 5 tasks. Each task/algorithm uses 20 trials.

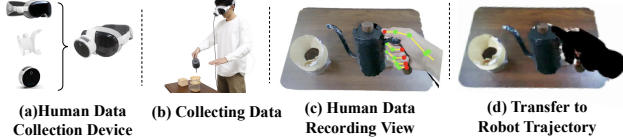


Figure 12. (a,b) show the components of UniDex-Cap. (c,d) shows the example captured data and converted robot-executable trajectories.

terous hands—together with FAAS—indeed enables zero-shot cross-hand skill transfer.

5.4. UniDex-Cap for Human-Robot Data Co-train

We introduce **UniDex-Cap**, a practical data-capture setup that records synchronized RGB-D streams and hand/head poses. The system combines an Apple Vision Pro for hand and head pose estimation, an Intel RealSense L515 for high-quality RGB-D, and a custom 3D-printed mount to physically couples the two sensors with a fixed rigid transform. This transform is calibrated to ensure the RGB-D stream and the hand/head poses are time-synchronized and expressed in the shared coordinate frame. As illustrated in Fig. 12, we then apply the human-to-robot transformation pipeline (Sec. 3.2) to convert captured human data into robot-executable trajectories. In addition, we perform a viewpoint transformation to align human and robot perspectives and downsample the human motion to match typical teleoperation speeds.

Leveraging UniDex-Cap, we collect human demonstrations, transform them, and *co-train* with real-robot data on *Make Coffee* task to quantitatively explore the effect of human demos during the finetuning stage. Figure 13 reports average task progress versus the numbers of co-trained transformed human demos (h) and robot demos. We observe: (i) **Retargeted human data helps, but robot data is indispensable**. Although for a fixed r , increasing h con-

sistently improves average task progress within our evaluated range but success always remains near zero without any robot data. (ii) **Human-robot exchange rate $\approx 2:1$** . From Fig. 13, the boundary separating the “high-performance” region (comparable to the $r=50$ robot-only result green area) has slope ≈ 2 , suggesting roughly *two human demos can substitute for one robot demo*. (iii) **Cost efficiency**. On *Make Coffee* task, human demos are $\sim 5.2\times$ faster to collect than real robot demos; considering the $\approx 2:1$ exchange rate, co-training with human demos can substantially reduce data collection cost.

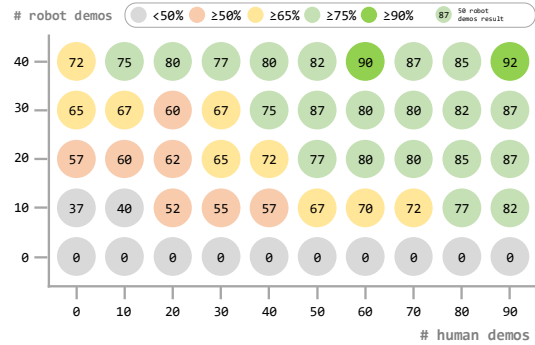


Figure 13. **Human-Robot co-training**. Average task progress versus the numbers of transformed human demos (h) and robot demos (r). Colors indicate different performance bands (green: comparable to the $r=50$ robot-only result). Each point averages over 20 trials.

6. Conclusion

We presented UniDex, a robot foundation suite built from egocentric human videos, comprising UniDex-Dataset, UniDex-VLA, and UniDex-Cap. We believe UniDex can serve as a practical foundation platform for the community, accelerating progress toward general, scalable, and transferable dexterous manipulation.

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